

A & B are similar for some M,

$$B = M^{-1}AM$$

Example,

$$S^{-1}AS = \Lambda$$

A is similar to  $\Lambda$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$\rightarrow \lambda = 3, 1$                        $\rightarrow \lambda = 3, 1$

$$\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -15 \\ 1 & 6 \end{bmatrix}$$

$\rightarrow \lambda = 3, 1$                        $\rightarrow \lambda = 3, 1$

$$M^{-1} \quad A \quad M = B$$

↳ Have same eigen-values, # eigen-vectors

Why?

$$Ax = \lambda x$$

$$B = M^{-1}AM$$

$$M^{-1}AMM^{-1}x = \lambda M^{-1}x$$

$$BM^{-1}x = \lambda M^{-1}x$$

↳  $\lambda$  stayed the same

↳ Eigen-vector of B is  $M^{-1}$  (eigen-vector of A)

Bad case  $\lambda_1 = \lambda_2$

let it be 4

$$\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} \leftarrow \text{Big family} \xrightarrow{M^{-1}AM} \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 17 & 4 \end{bmatrix}$$

Only member in its family  
 NOT same family  
 Jordan form  
 $N'DM = 0$

Jordan Block

$$J_i = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \lambda_i & 1 & \\ & & \ddots & 1 \\ 0 & & & \lambda_i \end{bmatrix}$$

Has one eigen-vector only

Example,

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

NOT similar

Jordan's Theorem:

Every square matrix  $A$  is similar to Jordan matrix  $J$

$$J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_d \end{bmatrix}$$

# blocks = # eigen-vectors

1. Good Case:

$J$  is  $\Lambda$